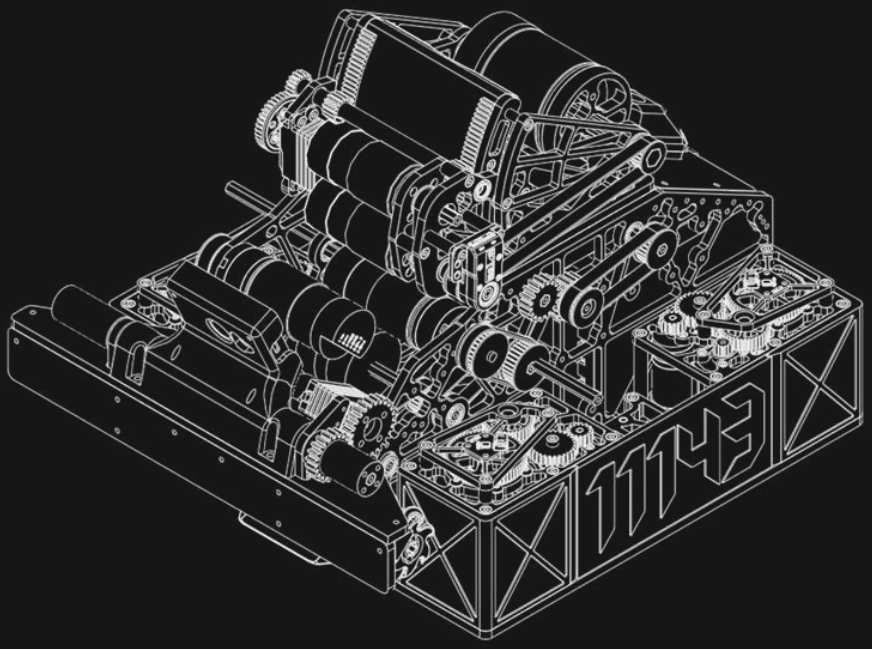
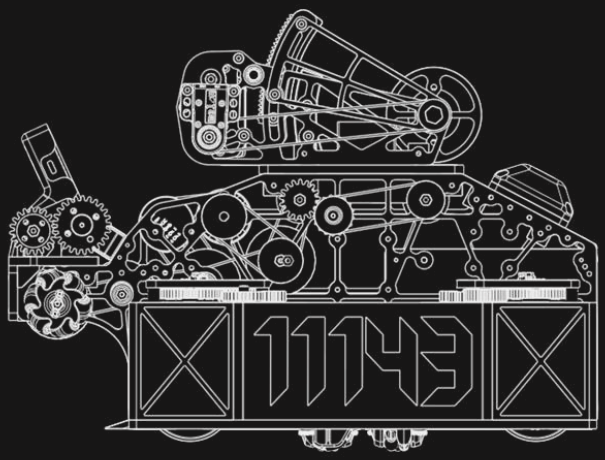
ENGINEERING PORTFOLIO

FTC - FIRST TECH CHALLENGE

BARKER YELLOWBACKS

11143





2026-2027

Sydney
Australia

FORMAT GUIDE

11143

COLOURS

- **Page** #171616
- **Text** #F2F2F2
- **Yellow** #F5E660
- **Grey** #C4C4C4
- **Pro / pass** #6FD98B
- **Con / fail** #FF7B72

TYPE

BANNER TITLE 19 PT BOLD CAPS

BOX HEADING 12 PT BOLD CAPS

Sub-heading 11.5 pt yellow

Body 10.5 pt Montserrat, with **key words in yellow**, **pros in green** and **cons in red**.

- Bullets are small yellow squares

Captions, tables and footer 10 pt grey (the smallest size: FTC A201)

Handwritten labels 15 pt

CHALLENGE TO SOLUTION

CHALLENGE

In simulation the chassis stopped aiming about **12 degrees** short: small wheel commands were inside the static friction dead zone.

SOLUTION

Added wheel feedforward based on battery voltage.

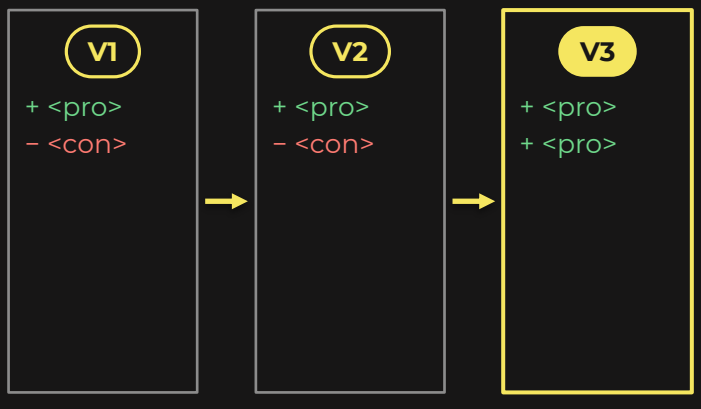
RESULT

Simulation aims on target. Real robot: <RESULT: aim error in degrees>.

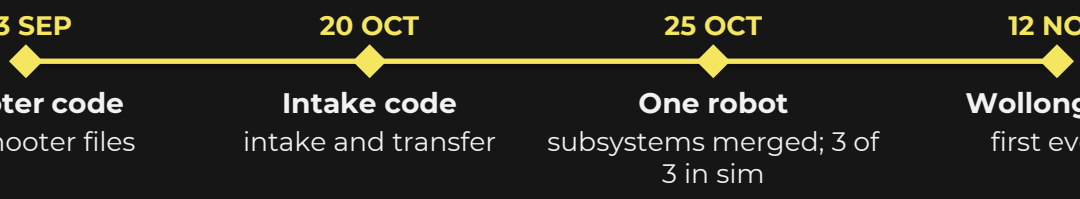
OPTIONS COMPARED

DRIVETRAIN	SCORE /25	STATUS
Swerve	<score>	CHOSEN
Mecanum	<score>	NOT CHOSEN
Tank	<score>	NOT CHOSEN

ITERATIONS



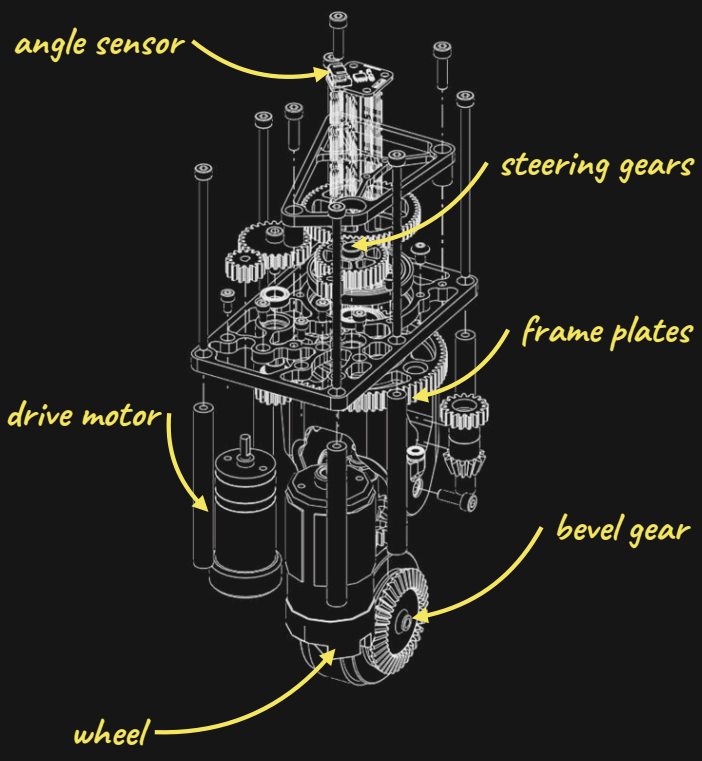
SEASON SO FAR (FROM OUR CODE HISTORY)



SWERVE DRIVETRAIN

11143

ONE SWERVE POD



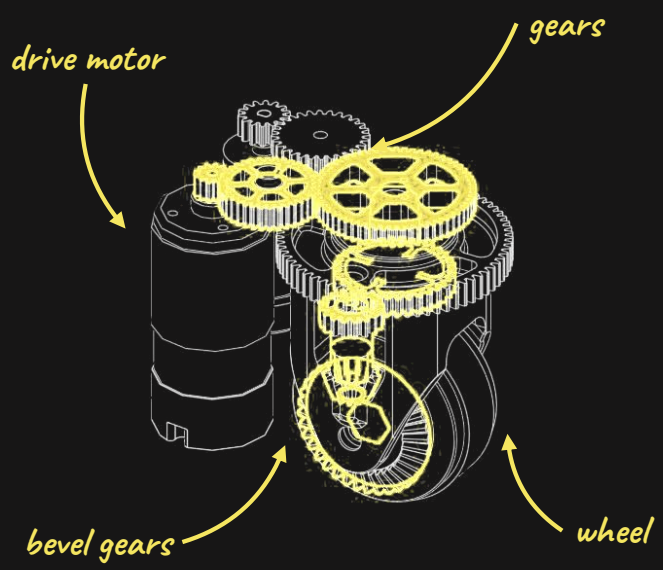
KEY FEATURES

- **4 pods**, each with a drive motor and a steering servo
- An **absolute angle sensor** in every pod, so each wheel knows where it points from the start
- Pod centres **239 x 239 mm** apart; wheels **60.9 mm**
- Robot **407.7 x 327.0 mm**, **290.9 mm** tall (CAD)

Why swerve, again

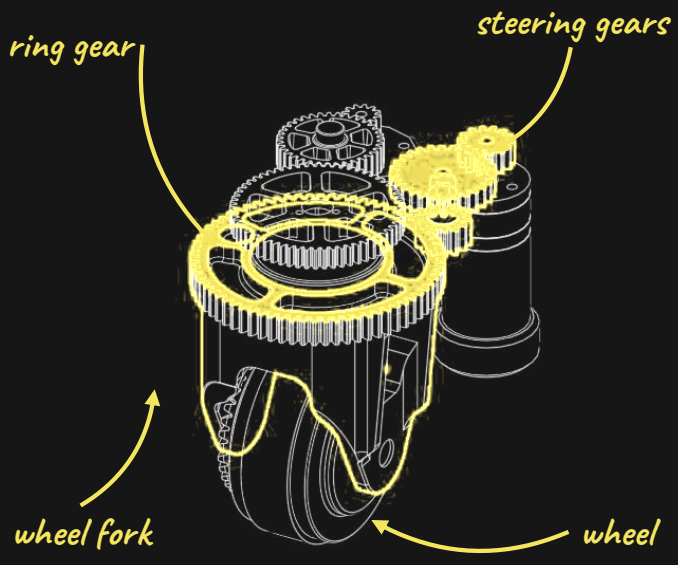
- **X-lock:** the wheels turn inwards, so defence cannot push us off our spot
- Full-grip wheels: more **speed and pushing power** than mecanum
- **Aim while driving:** drive one way, face the goal
- We built swerve last season, so the risk was low

WHEEL DRIVE



Motor, gears, then a bevel gear turn the wheel

MODULE ROTATION



Servo and ring gear turn the whole pod

TESTS

TEST	RESULT	STATUS
Top speed	<RESULT: m/s>	**NOT TESTED**
Steering angle error	<RESULT: degrees>	**NOT TESTED**

SHOOTER DESIGN

WHAT WE TRIED

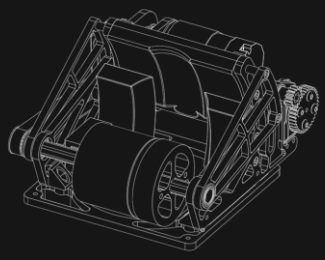
TURRET, LAST SEASON

+ aims without turning the robot
− months of tuning (Apr-Jun 2026)

TURRET, THIS SEASON

+ what we wanted first
− weight and space
− 1 more motor or servo
− no time to build it

FIXED REAR SHOOTER



+ fewer parts
+ shoots out of the flowers without turning
− the whole robot must turn to aim

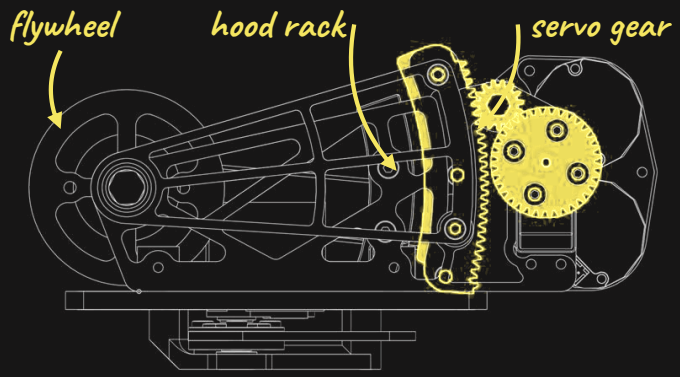
WHAT WE WANTED, AND WHY

- **Two flywheels:** more flywheel means **less waiting** between shots
- **Moving hood:** shoot from anywhere on the field
- **Facing backwards:** balls collect along the wall, so we drive up to the wall and aim, instead of collecting and then turning round

What we changed

We drew and coded a turret, but never built it. On **12 October** we removed it: the swerve base turns the robot to aim instead (auto-aim, page 4).

HOOD DRIVE



A servo gear drives a rack to tilt the hood

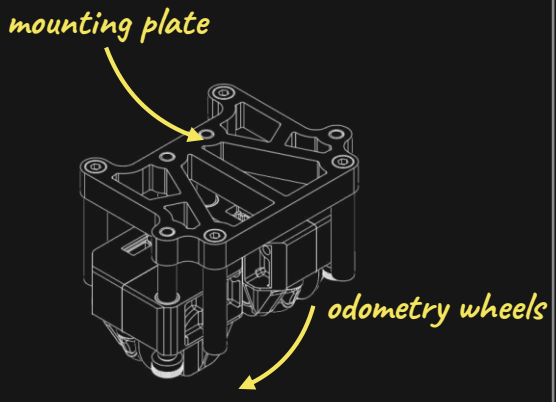
TESTS

TEST	RESULT	STATUS
Shots in from 3 distances	<RESULT: in / 10>	**NOT TESTED**
Time between shots	<RESULT: s>	**NOT TESTED**

WHAT EACH PART DOES

PART	PROBLEM IT SOLVES	WHAT IT DOES
4 drive motors	moving in any direction	drive each swerve wheel
4 steering servos + 4 angle sensors	pointing every wheel the right way	turn each pod and read its angle
IMU	knowing which way we face	heading for driving and aiming
3 beam sensors	knowing how many balls we hold	sense balls at bottom, middle and top of the intake
Gate servo	balls reaching the shooter too early	holds balls until the shot is ready
2 shooter motors	the same shot every time	flywheel held at target speed (PIDF)
Hood servo	shooting from different distances	sets the launch angle from our shot table
Voltage and current checks	brown-outs and stalled motors	logged every loop

ODOMETRY POD



In CAD; the code uses wheels + IMU for now

AUTO-AIM

CHALLENGE

We dropped the turret, so the shooter is fixed facing backwards.

SOLUTION

Holding aim turns the whole robot until it is within **2.5 degrees** of the goal.

RESULT

Simulation: **3 of 3** scored, also while moving.
Robot: <RESULT: shots in / 10>.

MEET THE TEAM

OUR TEAM

EB — **Ethan B.**
LEAD DESIGN
assembly

BL — **Brandon L.**
CAD
assembly

IH — **Isaac H.**
LEAD CODE
CAD, assembly

RT — **Ryan T.**
DRIVER
CAD

WQ — **Will Q.**
LEAD STRATEGY
drive coach

IW — **Ivy W.**
PROGRAMMING
HR

TX — **Tina X.**
CODE
assembly

CW — **Chelsea W.**
CODE
assembly

CL — **Chloe L.**
ELECTRICAL
assembly

CS — **Chloe S.**
STRATEGY
code, assembly

CC — **Cooper C.**
SCOUTING
assembly

JC — **Jemima C.**
PROGRAMMING
scouting

LT — **Lucas T.**
CAD
assembly

OUTREACH

- **7 members** teach robotics to Darkinjung students once a fortnight (<number> students since <month>)
- Ethan B. and Isaac H. mentor <number> VEX teams
- Ethan B., Isaac H. and Brandon L. volunteered at the Barker VEX competition (<date>)
- We ran the robotics part of the Round Square International Conference at Barker (<date>, <number> people)

Endnote: text drafted and laid out with help from an AI assistant. All results, photos, CAD, code and decisions are the team's own; the team checked every page.